Matthew Hamilton

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Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science in Systems Design Engineering *Sep. 2026 – May 2031 (Expected)*

- **Scholarships**: Alumni @ Microsoft Engineering Scholarship (1 recipient/year, facultywide)
- **Relevant Coursework**: Digital Computation (SYDE 121, C++), Linear Algebra (MATH 115)

Projects

LaserTagNow | *Swift, SwiftUI, ARKit, NearbyInteraction, Network Framework* 2026

- Won $1,200 + 2nd Place at SummerHacks
- Built a fully serverless, peer-to-peer multiplayer game for iOS supporting real-time multi-player matches with zero backend infrastructure, by designing a frame-independent networking protocol that encodes spatial events (hits, drops, positions) as affine combinations of player coordinates, letting each device resolve state into its own local AR frame without a shared world map
- Achieved reliable, real-time target-lock accuracy from noisy ∼5Hz Ultra-Wideband bearing data, by fusing NearbyInteraction ranging with ARKit camera pose estimation into a custom aim-assist algorithm balancing responsiveness against false locks (e.g., through walls)
- Shipped a complete real-time game loop — lobbies, combat, power-ups, death/respawn, and match summaries — by architecting an idempotent, order-tolerant state-sync layer that kept UI stable despite late or dropped peer-to-peer messages

Isle | *Swift, SwiftUI, AppKit, Core Audio, Core Animation, macOS* 2026 – Present

- Shipped a native macOS app that turns the MacBook notch into a live island for music and coding-agent sessions, rendering over the camera housing without ever taking focus or resizing the window beneath the cursor
- Cut CPU 91% while animating (19.95% → 1.88%) by profiling the render path and moving the island's animations off SwiftUI onto Core Animation layers
- Integrated the island with Claude Code as a live session monitor that opens on its own when an agent needs input, and stays correct across concurrent sessions
- Drove an audio-reactive waveform that follows the music itself rather than system volume, falling back to a procedural pattern when audio permission is declined rather than failing closed

Retermina | *Tauri v2, Rust, React, TypeScript* 2026 – Present

- Delivered a local-first, Claude-centric desktop terminal workspace by building on Tauri v2 (Rust + React/TypeScript), where a Rust backend spawns and multiplexes native PTY sessions over Tauri IPC with tools docked in a Cursor-style drawer
- Kept terminals — and the dev servers inside them — alive across app quit, crash, or auto-update by architecting a live-session layer that hosts every PTY in a separate re-exec'd process, reconnecting over a Unix-domain socket and replaying scrollback on relaunch, with a grace-TTL reaper reaping lingering daemons
- Turned it into a supervision layer for coding agents: embedded Claude Code with a live context-window gauge (parses session JSONL to chart context fill against the model window), per-project token and cost accounting, and cross-restart session resume that reconnects the agent and rehydrates its transcript
- Shipped universal (Apple Silicon + Intel) macOS releases via a GitHub Actions pipeline and a minisign-signed in-app auto-updater that verifies each download against a `latest.json` manifest before applying

Technical Skills

Languages: Swift, TypeScript, Rust, Python, C++, JavaScript, HTML/CSS
Frameworks & Libraries: SwiftUI, AppKit, ARKit, Core Audio, Core Animation, React, Node.js, Tauri v2
Tools & Platforms: Git, GitHub Actions, Xcode, Instruments, Linux, Docker, Vite, pytest
Areas: Real-time systems, peer-to-peer networking, native desktop & mobile development, sensor fusion, performance profiling